

**NAME**

saw – get the standard atomic weight

**LIBRARY**

ciaaw library (*libciaaw*, *-lciaaw*)

**SYNOPSIS**

```

Fortran      function saw(s,ab,u)result(res)
               character(len=*), intent(in) :: s
               logical, intent(in), optional :: ab
               logical, intent(in), optional :: u
               real(dp) :: res

C            double ciaaw_saw(char *s,int n,bool ab,bool u);

Python      saw(s:str,abridged:bool=True,uncertainty:bool=False)->float:

```

**DESCRIPTION**

Get the standard atomic weight for the element *s*. The default behavior is to provide the abridged value, *ab* is *TRUE*, recommended by the CIAAW. The uncertainty can be retrieved by setting *u*=*TRUE*.

Arguments:

*s*            Element symbol.  
*ab*           Set to False if the abridged value is not desired. Default to *TRUE*.  
*u*            Set to True if the uncertainty is desired. Default to *FALSE*.

**RETURN VALUE**

NaN           If the provided element is incorrect.  
-1            If the element does not have a SAW.

**EXAMPLES**

Fortran:

```

use ciaaw
saw("H", ab=.false., u=.true.)

```

C:

```

include <stdio.h>
include "ciaaw.h"
ciaaw_saw("H", 1, false, true)

```

Python:

```
import pyciaaw
pyciaaw.saw("H", ab=False, u=True)
```

**SEE ALSO**

***ciaaw*(3), *ciaaw\_cli*(1), *ciaaw\_version*(3), *ciaaw\_saw*(3), *ciaaw\_ice*(3), *ciaaw\_nice*(3), *ciaaw\_naw*(3), *ciaaw\_nnaw*(3), *ciaaw\_ices*(3), *ciaaw\_naws*(3), *ciaaw\_s2z*(3)**